

PRODUCTION AI ENGINEERING · T4 · REAL VS SIMULATED REAL VS SIMULATED

T4 Real vs Simulated: AI Control Plane

What in the POC is real code really run, what is simulated, what a production deployment would substitute, and what was not exercised.

T4

T4 · Production AI Engineering

4 min read · Published 2026-10-03 · Real vs simulated

The POC intentionally compresses production components into a small deterministic implementation so it can isolate the architectural property under test. Every row below is one of three things: **REAL** (the mechanism itself ran, and a proof check shows it), **SIMULATED** (a faithful local substitute for a system the property does not depend on), or **NOT EXERCISED** (argued in the editions, not tested here). The **PRODUCTION SUBSTITUTE** column says what replaces each compressed piece.

REAL

Component	Where	What ran	Proof check	Production substitute
Separate, long-lived runtime processes	acp/runtime/worker.py , acp/harness.py	real OS subprocesses speaking JSON lines; 16 across the run, 0 restarted; every response carries its process id, recorded as a label (rt-a/pid-1)	T4- R13- C05 , T4-R2- C03	the agent runtime fleet (containers, serverless workers) running the same SDK
Control-plane versioning, pointer, rollout, rollback	acp/controlplane/store.py	immutable vN.json bundles; 34 versions published across the run	T4-R8- C04	a versioned policy and configuration store with staged rollout
Bundle signing and verification	acp/common.py , acp/runtime/services.py	HMAC-SHA256 over each bundle's sha256; 34 signatures verified	T4- R13- C03 , T4-R9- C05	asymmetric signatures, keys in a KMS or HSM; the runtime holds only the public key
Change governance	acp/controlplane/admin.py , validate.py	scopes, restricting vs widening, two-person rule, break-glass, plaintext-credential rejection	T4- R12- C01 ... C08	an IdP-backed change workflow with MFA and just-in-time elevation
Hash-chained change log and runtime audit	acp/common.py	every row carries the previous row's hash; 14 audit and 14 change-log chains recomputed	T4- R13- C01 , T4- R13- C02	an append-only external audit store (WORM, transparency log); this is a tamper-evident local log under POC assumptions
Decision function and enforcement points	acp/runtime/pdp.py , sdk.py	every tool and model call decided against the current bundle, with the version and rule named	T4-R1- C02 , T4-R1- C03	a policy engine (OPA/Cedar-style) embedded in the runtime SDK and the MCP gateway
Last-known-good cache and failure policy	acp/runtime/sdk.py	staleness bound, fail-closed mutations, rejected bundles	T4-R9- C02 , T4-R9- C03	the same semantics, with staleness in seconds and freshness/revocation signals
Agent code and request hashing	acp/common.py , acp/experiments.py	sha256 over acp/agents/*.py before, at import and after; request hash of every run; 93 of 93 recomputed	T4- R13- C06 , T4- R13- C08	build provenance (SBOM, signed images) and request tracing
Architecture rules	tests/test_architecture.py , tests/test_invariants.py , tests/test_cheating.py	agents import nothing and name no model; the runtime cannot reach the control plane's mutation API; execution only after a decision; P2 fails if code, process or request changes	make test	the same rules in CI

SIMULATED

Component	Simulation	Production substitute
Agents' reasoning	fixed deterministic plans, no LLM, in the recorded run. The optional live mode (<code>make live</code> , L1–L3) lets a self-hosted model (Ollama: <code>qwen3:8b</code>) plan the incident agent's steps; not exercised in this evidence, illustrative only	a model-driven agent; the governance path is unchanged
Enterprise systems and MCP servers	<code>acp/systems.py</code> : deploy, billing, support, observability as in-process systems of record; side effects in <code>effects.json</code> , every call in <code>log.jsonl</code>	real services behind MCP servers with OAuth resource-server checks
Model endpoints	<code>ModelGateway</code> : deterministic text, token counts and cost	a model gateway enforcing the resolved model, data class and residency
Distribution channel	a shared directory read at each enforcement point	push or watch with acknowledgements, propagation delay and bundle expiry
Network partition, lagging replica, in-transit tampering	flags in <code>state/network/<instance>.json</code> (INJECTED)	real faults; chaos testing
Credential broker	in-process minting of audience-bound, short-lived credentials	a secrets manager issuing leased dynamic credentials to authenticated workloads; the control plane holds <code>secret://</code> references, never values
Approval service	a JSON store with eligibility and expiry	the durable approval gate from T3
Spend meter	a shared JSON file	a budget service with atomic counters
Time	logical ticks	wall-clock time; staleness in seconds
Administrators	named principals in <code>config/admins.yaml</code>	IdP groups, MFA, just-in-time elevation

NOT EXERCISED

Not exercised	Why it matters	Status in the Evidence Check
Real MCP protocol traffic and interoperability	P6 shows central revocation at a simulated boundary, not a working MCP deployment	Not tested (T4-C20)
Environment and tenant isolation, admin MFA, control-plane HA, backup and restore, real key management	the control plane becomes the most privileged system in the estate	Not tested (T4-C21)
Latency, throughput, availability	local decisions against a cached bundle are a design argument here, not a measurement	Argued (T4-C19)
Scale beyond three agents	the embedded baseline's edit counts are for three agents	Argued (T4-C18)
Real model behaviour	the recorded run has no model; the live mode is illustrative	not part of this evidence

Limitations

The POC intentionally compresses production components into a small deterministic implementation so it can isolate the architectural property under test. That choice buys exact replay (EXACT: a rerun from source differs only in masked process ids) and a clean causal story for P2, at the price of everything in the two tables above: the systems, models, distribution and time are simulated, and nothing here measures performance or proves a model will behave.

Evidence

Method

AI Control Plane

PRODUCTION AI ENGINEERING · THE SERIES SO FAR

FOUNDATION

F1 · MCP Tool Sprawl

Your AI agent has 500 MCP tools. Now what?

FOUNDATION

F2 · Layered Architecture

Your agent works in a demo. Why does it break in production?

FOUNDATION

F3 · Headless AI

Your AI shouldn't live inside the UI

TRUST

T1 · Agent Identity

Who is acting, and on whose authority?

TRUST

T2 · Authorization & Policy

What is this agent actually allowed to do?

PREVIOUS

T3 · Human-in-the-Loop

The Human-in-the-Loop chapter

CURRENT

T4 · AI Control Plane

Who governs every agent, and how does one change reach all of them?

NEXT

T5 · Observability & Governance

Your AI agent did something in production. Can you explain exactly what happened?

The full learning map (Start Here)

T4 · Production AI Engineering

Production AI Engineering · T4 · Real vs simulated. Every measured number here is substituted at build time from `control_plane_poc/runs/2026-10-03-recorded/facts.json`; the enterprise systems are simulated, every runtime is a real long-lived process, and the run is deterministic. Figures carry their provenance.

- Medium edition Technical deep dive Evidence Check POC README Proof cards
- Run summary Lab Console QA report